EXPERIMENT 6
FORENSIC ANALYSIS OF EVIDENTIAL IMAGE USING FTK

Date: 25/02/2026

AIM

Using **FTK** acquire an image and perform static analysis to mount an image of a drive

THEORY

FTK Imager is a digital forensic software tool developed by AccessData. It is widely used in computer forensics to acquire and analyze digital evidence from various sources, such as hard drives, removable media, and forensic images.

FTK Imager provides a range of features and capabilities that assist forensic examiners in their investigations. Some of its key features include:

1. **Disk Imaging**: FTK Imager allows forensic professionals to create forensic images or exact copies of digital storage media. These images can be analyzed without altering or tampering with the original evidence.
2. **File Analysis**: The software enables examiners to view and analyze the files and data within a forensic image. It provides various tools for searching, sorting, filtering, and examining file metadata and content.
3. **Mounting Images**: FTK Imager allows forensic images to be mounted as a logical drive, making it easier to access and analyze the content using standard file explorer interfaces.
4. **Keyword Searching**: The tool offers advanced keyword searching capabilities, which help investigators locate specific files, text strings, or metadata within a forensic image or a collection of images.
5. **Data Carving**: FTK Imager includes data carving functionality, allowing the recovery of deleted or hidden files by searching for file signatures and other data patterns.

6. **Reporting**: The software supports generating comprehensive reports summarizing the findings of a forensic examination. These reports can be customized and exported in various formats for use in legal proceedings or documentation.

FTK Imager is commonly used in conjunction with other digital forensic tools and software suites. It is designed to provide a user-friendly interface while offering powerful capabilities for forensic analysis and evidence acquisition.

PROCEDURE

I. Take image of the Disk

To perform disk imaging using FTK Imager, follow these general steps:

1. **Launch FTK Imager**: Start by launching the FTK Imager software on your computer. It is available for Windows and can be downloaded from the AccessData website.

2. **Select the Source**: In the FTK Imager main window, click on the "File" menu and choose the "Create Disk Image" option. A dialog box will appear where you can select the source device or media you want to image. This can be a physical hard drive, a removable storage device, or a forensic image file.

3. **Choose Destination**: After selecting the source, choose the destination where you want to save the disk image. This should be a separate storage device or location to ensure that the image does not overwrite or modify the original evidence.
4. **Select Imaging Options**: FTK Imager offers various imaging options that can be configured based on your requirements. These options include compression, encryption, and verification settings. You can choose to compress the image to save disk space, encrypt it for security, and enable verification to ensure data integrity.
5. **Start the Imaging Process**: Once you have configured the imaging options, click on the "Start" button to begin the imaging process. FTK Imager will create a forensic image of the selected source and save it to the specified destination.
6. **Monitor the Progress**: During the imaging process, FTK Imager will display the progress and provide information about the current status, including the number of sectors processed

and the estimated time remaining. You can monitor this information to track the progress of the imaging process.

7. **Verify the Image**: After the imaging process is complete, it is crucial to verify the integrity of the disk image. FTK Imager can perform a verification process to ensure that the created image is an exact replica of the source device. You should compare the hash values of the original source and the created image to ensure data integrity.
8. **Save a Report**: FTK Imager allows you to generate a report summarizing the imaging process and the details of the created image. You can save this report for documentation purposes and future reference.

II. Analyse the Image

To analyze a disk image using FTK Imager, you can follow these general steps:

1. **Launch FTK Imager**: Start by launching FTK Imager on your computer.
2. **Open the Disk Image**: In the FTK Imager main window, click on the "File" menu and select the "Add Evidence Item" option. Browse to the location where the disk image is saved and select it. FTK Imager will load the disk image for analysis.
3. **Explore the File System**: Once the disk image is loaded, you can explore the file system within the image. FTK Imager displays the directory structure and files contained in the image. You can navigate through the folders and files to examine their properties and contents.
4. **View File Metadata**: FTK Imager provides detailed metadata information for each file in the disk image. You can select a file and view its metadata, including file name, size, creation/modification dates, and other relevant information.
5. **Search for Files**: FTK Imager offers powerful search capabilities to locate specific files or information within the disk image. You can use keywords, file extensions, or other criteria to perform searches. The search results will be displayed, allowing you to further analyze the identified files.
6. **Extract Files**: If you need to extract specific files from the disk image, you can use FTK Imager to do so. Select the files you want to extract and choose the "Extract" option from the menu. Specify the destination folder where you want to save the extracted files.

7. **Analyze File Content**: FTK Imager allows you to analyze the content of individual files within the disk image. You can view text-based files, images, documents, and other file types supported by the software. This can help you extract valuable information and insights from the evidence.
8. **Generate Reports**: FTK Imager enables you to generate reports summarizing the analysis of the disk image. You can customize the report format, include relevant details, and export it for documentation purposes or legal proceedings.

III. Mount the Image

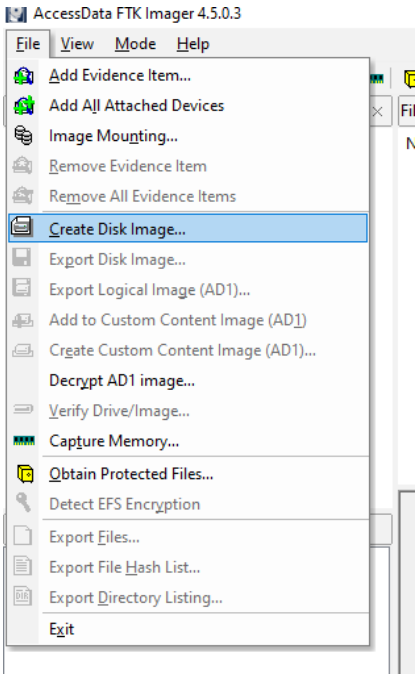
To mount a disk image using FTK Imager, you can follow these steps:

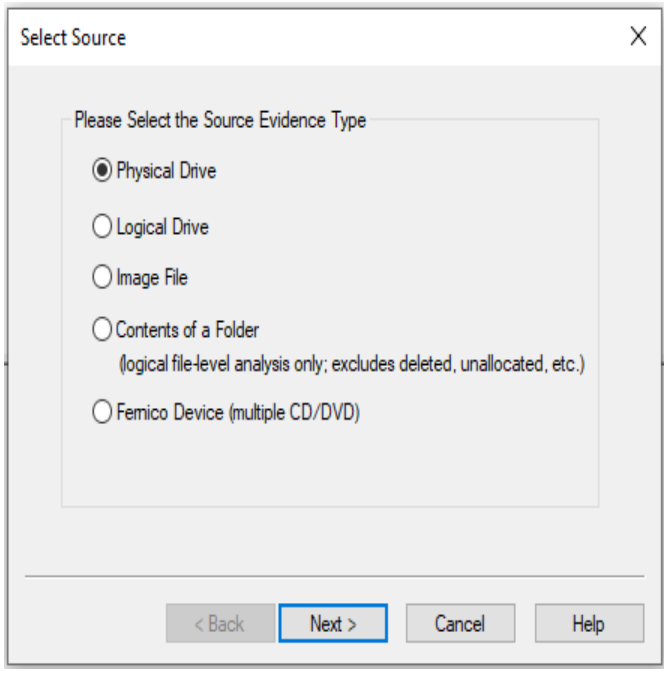
1. **Launch FTK Imager**: Start by launching FTK Imager on your computer.
2. **Open FTK Imager**: In the FTK Imager main window, click on the "File" menu and select the "Add Evidence Item" option.
3. **Select the Disk Image**: In the "Add Evidence Item" dialog box, browse to the location where the disk image is saved and select it. Click on the "Finish" button to load the disk image into FTK Imager.
4. **Mount the Disk Image**: Once the disk image is loaded, right-click on the disk image entry in the left-hand pane of FTK Imager. From the context menu, select the "Mount" option.
5. **Choose the Mount Options**: A dialog box will appear with mount options. Here, you can specify the drive letter you want to assign to the mounted disk image. You can also choose whether to mount the image as read-only or with write access.
6. **Click "OK" to Mount**: After configuring the mount options, click on the "OK" button to initiate the mounting process. FTK Imager will mount the disk image and assign it the specified drive letter.
7. **Access the Mounted Image**: Once the disk image is successfully mounted, you can access it like any other logical drive on your computer. Open Windows File Explorer, and you will see the mounted image listed as a separate drive with the assigned drive letter.
8. **Analyze and Work with the Mounted Image**: You can now navigate through the mounted disk image using File Explorer or any other file management tool. You can view and open files, analyze their content, perform searches, and conduct further analysis on the mounted image.

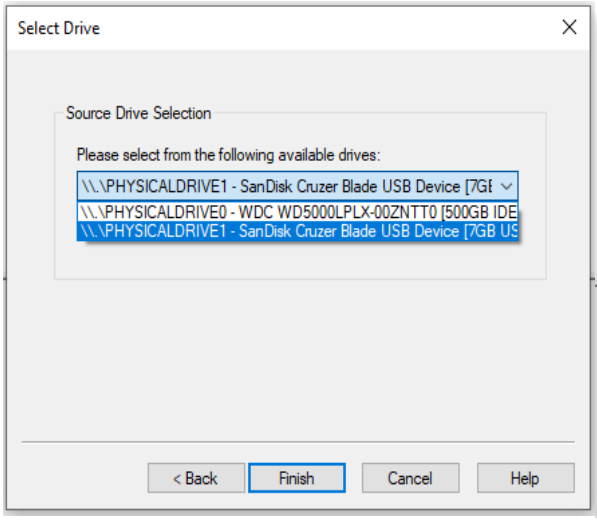
9. **Unmount the Image**: When you have completed your analysis or no longer need the mounted image, it is important to unmount it properly. Right-click on the mounted image in the left-hand pane of FTK Imager and select the "Unmount" option from the context menu.

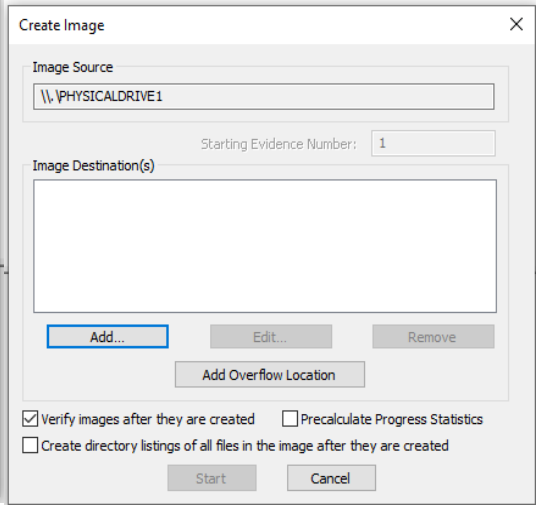
SCEENSHOTS

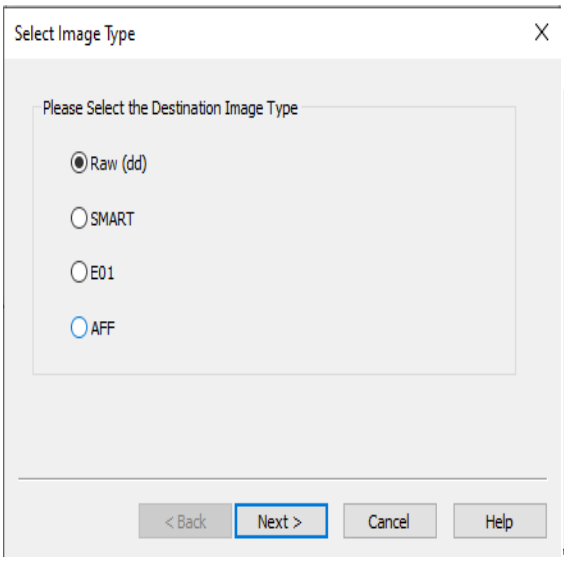
1. IMAGING A DISK







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SMART
E01
AFF
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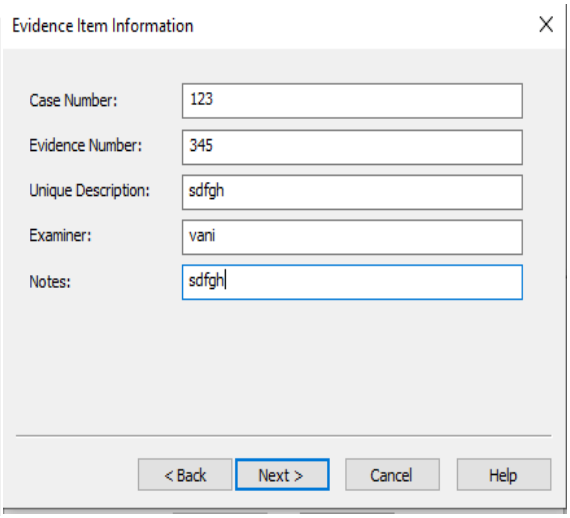
Next >
Cancel
Help


Evidence Item Information
Case Number:
123
Evidence Number:
345
Unique Description:
sdfgh
Examiner:
vani
Notes:
sdfgh
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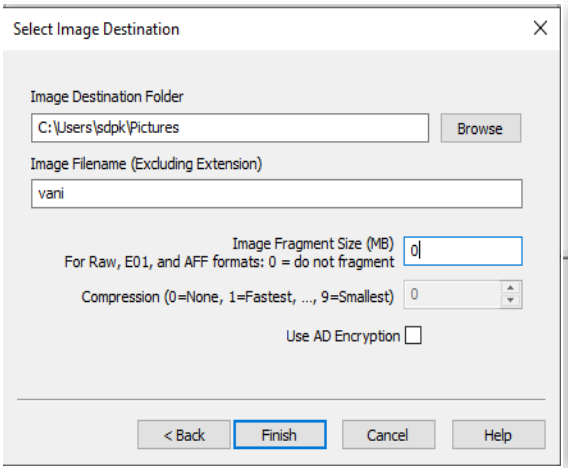
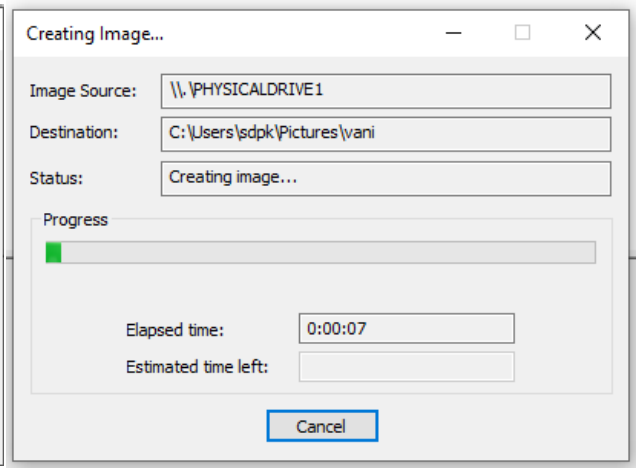
Next >
Cancel
Help


Select Image Destination
Image Destination Folder
C:\Users\sdpk\Pictures
Browse
Image Filename (Excluding Extension)
vani
Image Fragment Size (MB)
For Raw, E01, and AFF formats: 0 = do not fragment
0
Compression (0=None, 1=Fastest, ..., 9=Smallest)
0
Use AD Encryption
< Back
Finish
Cancel
Help


Creating Image...
Image Source:
\\.\PHYSICALDRIVE1
Destination:
C:\Users\sdpk\Pictures\vani
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Creating image...
Progress
Elapsed time:
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Estimated time left:
Cancel

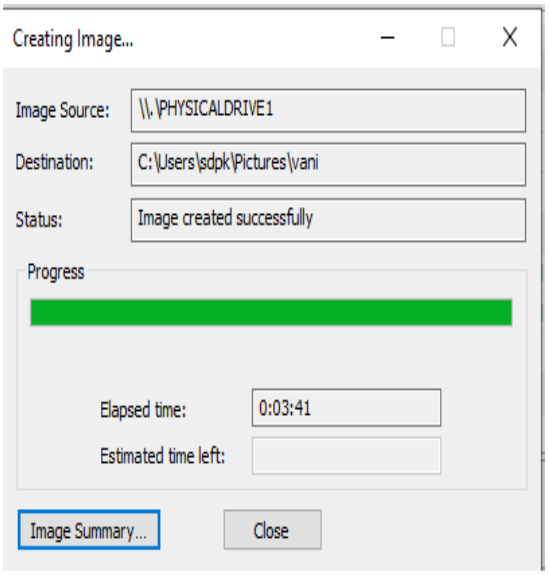
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Image Source:
\\.\PHYSICALDRIVE1
Destination:
C:\Users\sdpk\Pictures\vani
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Progress
Elapsed time:
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Estimated time left:
Image Summary...
Close

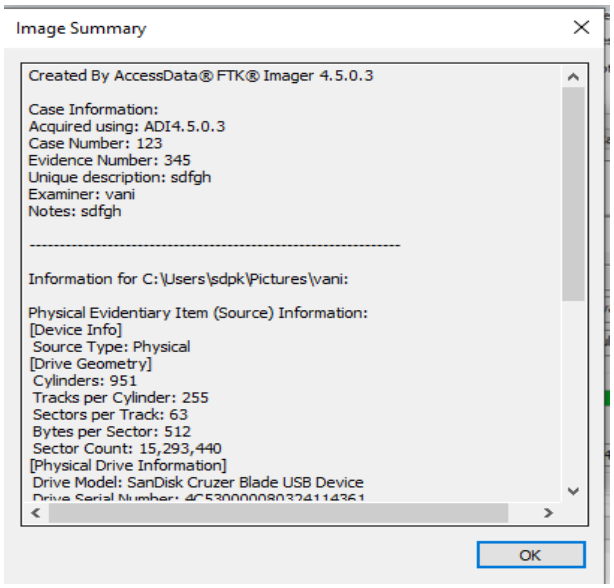
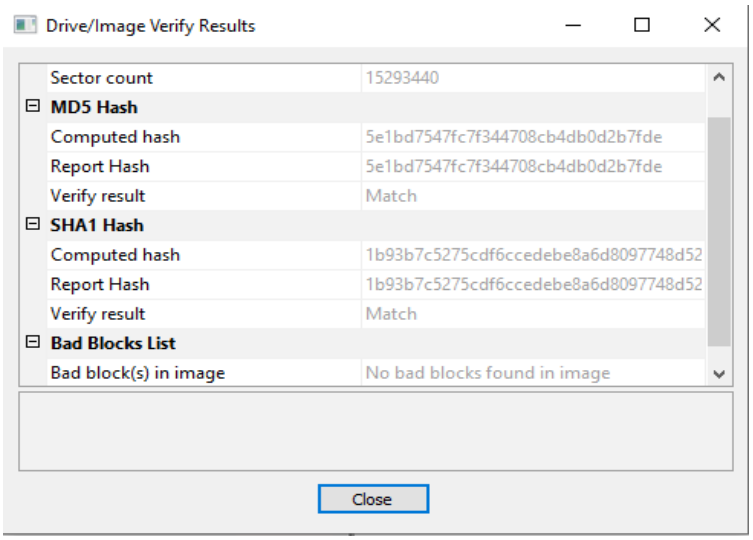


Image Summary
Created By AccessData® FTK® Imager 4.5.0.3
Case Information:
Acquired using: ADI4.5.0.3
Case Number: 123
Evidence Number: 345
Unique description: sdfgh
Examiner: vani
Notes: sdfgh

Information for C:\Users\sdpk\Pictures\vani:
Physical Evidentiary Item (Source) Information:
[Device Info]
Source Type: Physical
[Drive Geometry]
Cylinders: 951
Tracks per Cylinder: 255
Sectors per Track: 63
Bytes per Sector: 512
Sector Count: 15,293,440
[Physical Drive Information]
Drive Model: SanDisk Cruzer Blade USB Device
OK


Drive/Image Verify Results
Sector count
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MD5 Hash
Computed hash
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Report Hash
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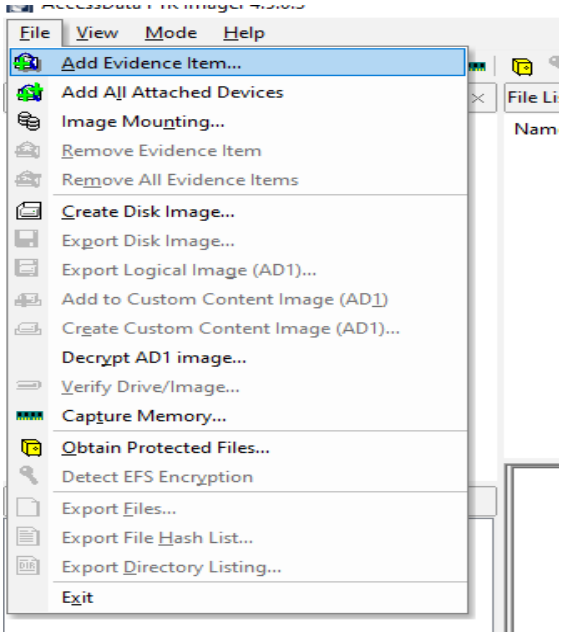
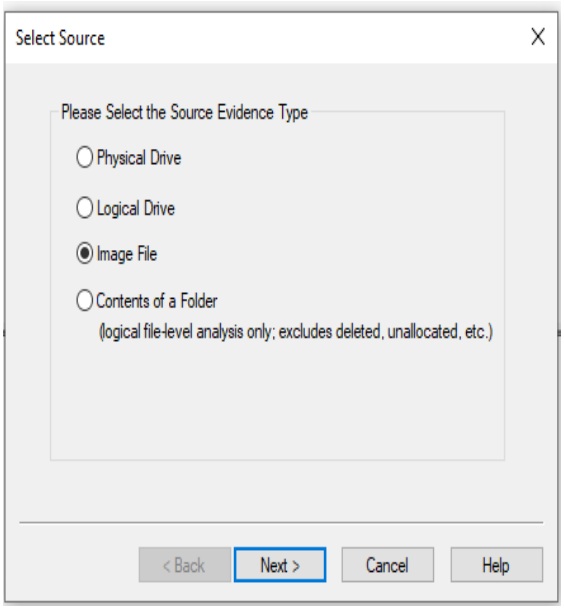
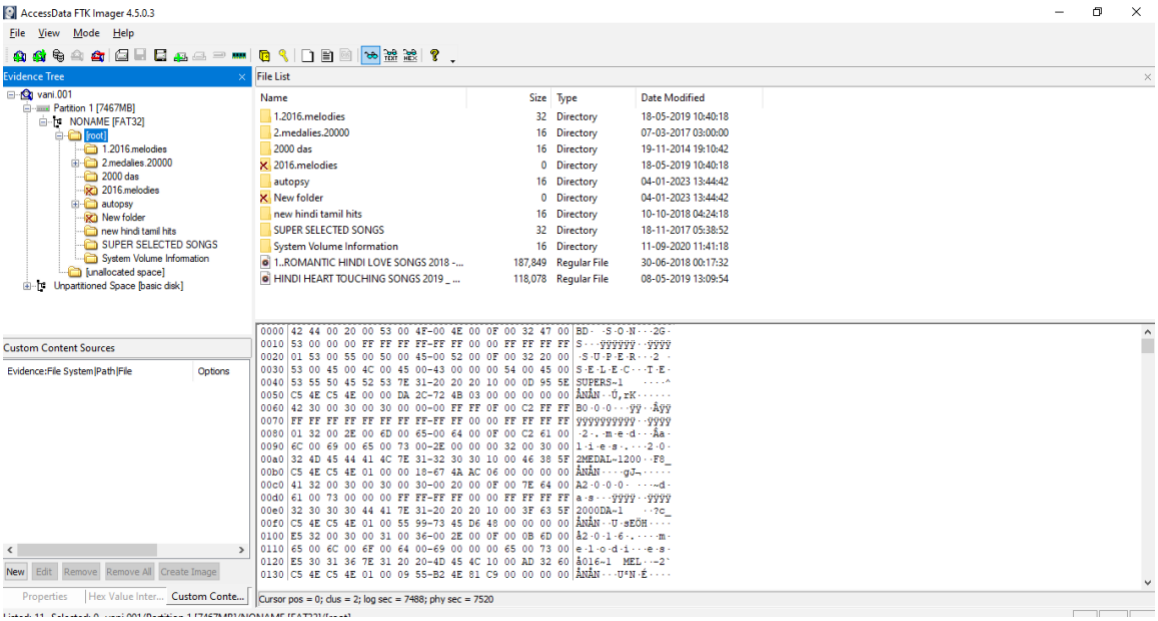
Verify result
Match
SHA1 Hash
Computed hash
1b93b7c5275cdf6ccedebe8a6d8097748d52
Report Hash
1b93b7c5275cdf6ccedebe8a6d8097748d52
Verify result
Match
Bad Blocks List
Bad block(s) in image
No bad blocks found in image
Close

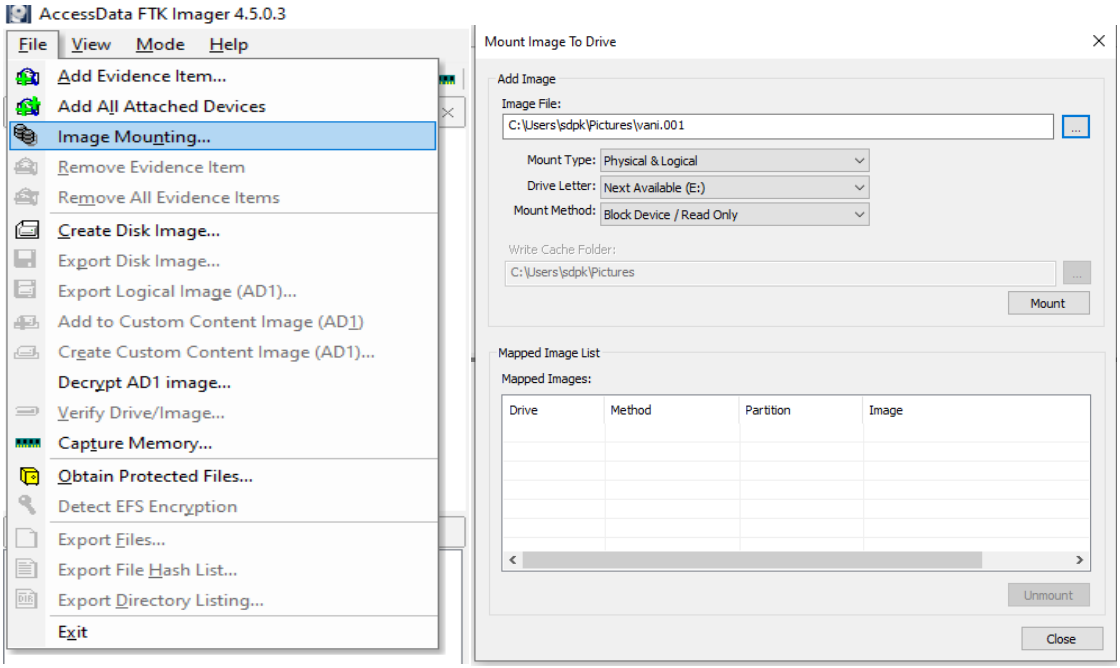
2. ANALYSING IMAGE

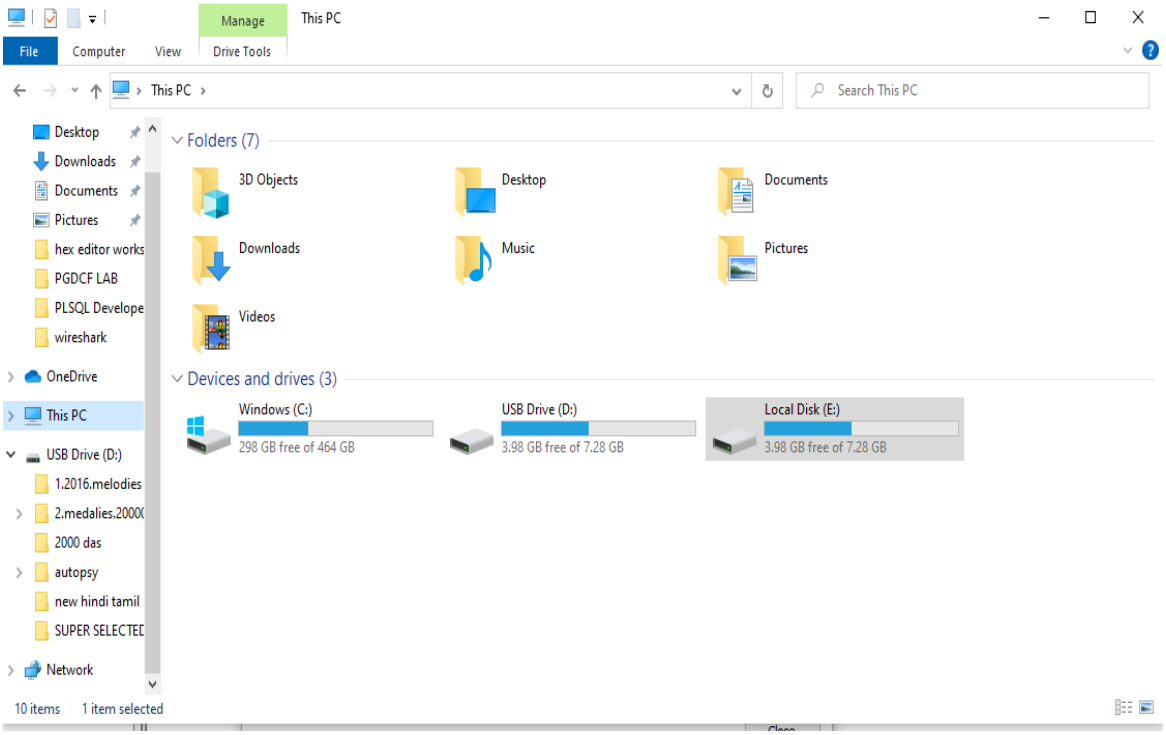






3. MOUNTING THE IMAGE





RESULT

Familiarised with the working of FTK Imager and obtained the information about the image, Image mounting is successfully completed.